



Assignment-2 (Set-1)

Course: B.Tech (CSE)

Year: III Semester: VI

Session: 2025-26

Subject Code & Name: BCSC-0011 (Theory of Automata & Formal Language)

Q1. For the below given CFG:

$S \rightarrow aAS \mid SS \mid a$

$A \rightarrow ba \mid SbA$

Find LMD, RMD and derivation tree for string "**aabbbaa**".

Q2. Eliminate the useless productions from the given grammar. Enlist all the steps clearly.

$S \rightarrow aaB \mid abA \mid aaS$

$A \rightarrow aA$

$B \rightarrow ab \mid b$

$C \rightarrow ad$

Q3. Determine whether the below grammar G is **ambiguous** or not.

$S \rightarrow a \mid Sa \mid SbS \mid SSb \mid bSS$

Q4. Convert the following Context Free grammar to equivalent CNF.

$S \rightarrow aAbB$

$A \rightarrow Ab \mid b$

$B \rightarrow Ba \mid a$

Q5. Convert the following CFG into equivalent GNF:

$S \rightarrow AB$

$A \rightarrow BS \mid b$

$B \rightarrow SA \mid b$



Assignment-2 (Set-2)

Course: B.Tech (CSE)

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Subject Code & Name: BCSC-0011 (Theory of Automata & Formal Language)

Q1. Suppose G be the grammar. Find LMD, RMD and Derivation Tree for string “00110101” :

$S \rightarrow 0B \mid 1A$

$A \rightarrow 0 \mid 0S \mid 1AA$

$B \rightarrow 1 \mid 1S \mid 0BB$

Q2. Eliminate the useless productions from the given grammar.

$S \rightarrow AB \mid AC$

$A \rightarrow aAb \mid bAa \mid a$

$B \rightarrow bbA \mid aaB \mid AB$

$C \rightarrow abCa \mid aDb$

$D \rightarrow bD \mid aC$

Q3. Suppose a tiny robot language encodes simple commands where the letter a means "step forward" and b means "beep". A valid program for the robot must end with a beep (b). The grammar G that generates robot programs is:

$S \rightarrow AB \mid aaB$

$A \rightarrow a \mid Aa$

$B \rightarrow b$

Determine whether grammar G is **ambiguous** or not.

Q4. Convert the following CFG to equivalent GNF:

$S \rightarrow YY \mid 0$

$Y \rightarrow SS \mid 1$

Q5. Design a Turing Machine to accept the language:

$L = \{ ww^r \mid w \in \{a, b\}^* \}$.



Assignment-2 (Set-3)

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Q1. Suppose G be the grammar given as:

$S \rightarrow aB \mid bA$

$A \rightarrow a \mid aS \mid bAA$

$B \rightarrow b \mid bS \mid aBB.$

For the string “**bbaababa**” find :

a) LMD b) RMD c) Derivation Tree

Q2. Eliminate null productions from the given grammar. Enlist all the steps clearly.

$S \rightarrow ABAC$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow bB \mid \epsilon$

$C \rightarrow d$

Q3. Determine whether the below grammar G is **ambiguous** or not.

$S \rightarrow a \mid abSb \mid aAb$

$A \rightarrow bS \mid aAAb$

Q4. Convert the following grammar into CNF:

$S \rightarrow ASA \mid bA$

$A \rightarrow B \mid S$

$B \rightarrow c$

Q5. Construct a TM for language $L = \{ a^{n+2} b^n \mid n \geq 1 \}$.



Assignment-2 (Set-4)

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Q1. Simplify the following grammar:

$S \rightarrow 0A0 \mid 1B1 \mid BB$

$A \rightarrow C$

$B \rightarrow S \mid A$

$C \rightarrow S \mid \epsilon$

Q2. Identify whether below grammar is **ambiguous** or not.

$S \rightarrow aS \mid a \mid \epsilon$

Q3. Convert the given CFG into their corresponding CNF:

$E \rightarrow E+E \mid E^*E \mid a \mid b \mid c$

Q4. Convert the following CFG to equivalent GNF:

$S \rightarrow A \mid AB$

$A \rightarrow aA \mid \epsilon$

$B \rightarrow bB \mid b$

Q5. Given a Context Free Grammar $G = ((S,A,B).\{0,1\}.P.S)$ with the production set as follows:

$S \rightarrow 0B \mid 1A$

$A \rightarrow 0S \mid 1AA \mid 0$

$B \rightarrow 1S \mid 0BB \mid 1.$

Design a PDA M corresponding to this CFG.



Assignment-2 (Set-5)

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Q1. Simplify the below grammar:

$S \rightarrow aAa$

$A \rightarrow Sb \mid bCC \mid DaA$

$C \rightarrow abb \mid DD$

$D \rightarrow aDA$

$E \rightarrow aC$

Q2. Determine whether the below grammar G is **ambiguous** or not.

$S \rightarrow S+S \mid S*S \mid id$

Q3. Convert the following Context Free grammar to equivalent CNF.

$S \rightarrow aAa \mid bBb$

$A \rightarrow C \mid a$

$B \rightarrow C \mid b$

$C \rightarrow CDE \mid \epsilon$

$D \rightarrow A \mid B \mid ab$

Q4. Find a PDA equivalent to the given grammar:

$S \rightarrow aSbb \mid a$

Simulate the designed PDA for the string **“aabb”**.

Q5. Design a Turing machine to compute the sum of two positive integers n and m , where $n, m > 1$.



Assignment-2 (Set-6)

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Q1. Simplify the following CFG (G).

$G = ((S, A, B, C), \{a, b\}, P, S)$

$P : S \rightarrow aAa, A \rightarrow bBB, B \rightarrow ab, C \rightarrow aB$

Q2. Determine whether the below grammar G is ambiguous or not.

$A \rightarrow AA \mid aAb \mid bAa \mid \epsilon$

Q3. Convert the following CFG to equivalent CNF:

$S \rightarrow aAD$

$A \rightarrow aB \mid bAB$

$B \rightarrow b$

$D \rightarrow d$

Q4. Find a PDA equivalent to the given grammar:

$S \rightarrow XY$

$X \rightarrow aX \mid \epsilon$

$Y \rightarrow bYc \mid \epsilon$

Q5. Construct a Turing machine to compute the given function:

$$f(n) = \begin{matrix} n-2 & \text{if } n \geq 2 \\ 1 & \text{if } n < 2 \end{matrix}$$



Assignment-2 (Set-7)

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Q1. Identify whether below grammar G is **ambiguous** or not.

$B \rightarrow aBbB \mid bBaB \mid \epsilon$

Q2. Construct the PDA from the given CFG.

$S \rightarrow 0SX \mid 1SY \mid \epsilon$

$X \rightarrow 1$

$Y \rightarrow 0$

Q3. If a language L and its complement L' are both recursively enumerable then prove that L is recursive.

Q4. Design a Turing Machine over alphabet { a, b, c } for

$L = \{ a^n b^n c^n \mid n \geq 1 \}$.

Q5. Convert the below given PDA into CFG:

a.) $\delta(q_0, a, Z_0) = (q_0, XZ_0)$

b.) $\delta(q_0, a, X) = (q_0, XX)$

c.) $\delta(q_0, b, X) = (q_1, \epsilon)$

d.) $\delta(q_1, b, X) = (q_1, \epsilon)$

e.) $\delta(q_1, \epsilon, Z_0) = (q_1, \epsilon)$



Assignment-2 (Set-8)

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Q1. Determine whether the below grammar G is **ambiguous** or not.

$S \rightarrow SS$

$S \rightarrow a$

$S \rightarrow b$

Q2. Construct the PDA from the given CFG.

$S \rightarrow XS \mid ^$

$A \rightarrow aXb \mid Ab \mid ab$

Q3. Design a TM to recognize the strings **(ab)⁺**.

Q4. Reduce the below grammar into Greibach Normal form:

$S \rightarrow A0$

$A \rightarrow 0B$

$B \rightarrow A0 \mid 1$

Q5. Construct a TM for $n \bmod 2$ where 'n' is represented in unary form on input tape.



Assignment-2 (Set-9)

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Q1. Determine whether the below grammar G is **ambiguous** or not.

$S \rightarrow aS \mid Sa \mid \epsilon$

Q2. Design a Turing Machine M to find:

a.) 1's complement of a binary number.

b.) 2's complement of a binary number.

Q3. Eliminate all the unit productions from the below grammar:

$S \rightarrow AB$

$A \rightarrow a$

$B \rightarrow C \mid b$

$C \rightarrow D$

$D \rightarrow E$

$E \rightarrow a$

Q4. Reduce the below grammar into Chomsky Normal form:

$S \rightarrow 0S0 \mid 1S1 \mid A$

$A \rightarrow 2B3$

$B \rightarrow 2B3 \mid 3$

Q5. Construct a Turing machine that accepts all strings starting with "01".



Assignment-2 (Set-10)

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Q1. Determine whether the below grammar G is **ambiguous** or not.

$$S \rightarrow aS \mid aSbS \mid \epsilon$$

Q2. Reduce the below grammar into Chomsky Normal form.

$$G = (\{ S \}, \{ a, b, c \}, \{ S \rightarrow a \mid b \mid cSS \}, \{ S \}).$$

Q3. Show that the below CFG is ambiguous:

$$S \rightarrow SS \mid (S) \mid \epsilon$$

Q4. Design a Turing Machine to reverse the binary string on the input tape.

Q5. Design a Turing Machine to identify any given input binary no. is even.

Q6. Convert the below given PDA into CFG:

a.) $\delta(q_0, a, Z_0) = (q_0, XZ_0)$

b.) $\delta(q_0, a, X) = (q_0, XX)$

c.) $\delta(q_0, b, X) = (q_1, \epsilon)$

d.) $\delta(q_1, b, X) = (q_1, \epsilon)$

e.) $\delta(q_1, \epsilon, Z_0) = (q_1, \epsilon)$